

Power PCB

Power PCB

Battery input

- **BAT+ (12V IN) → Fuse → Main Power Switch → SW_12V+ node (switched 12V+)**
- **BAT- (GND IN) → GND BUS (unswitched ground)**

Usual best practice: fuse and switch the **positive** lead only. Ground stays common.

Switched 12V distribution (the backbone)

From **SW_12V+ node**:

- → **Main 12V Rail Output (to Main PCB 12V IN)**
- → **5V Step-Down Converter VIN+**
- → **3.3V Step-Down Converter VIN+**
- → **48V Step-Up Converter VIN+**
- → **Battery Capacity Indicator + (sense/input)**

From **GND BUS**:

- → **Main PCB GND**
 - → **All converter VIN- / GND**
 - → **Battery indicator GND**
 - → **All output grounds (shared)**
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Converter wiring

1) Step-Down Converter (5V)

Input

- VIN+ → **SW_12V+**
- VIN- → **GND BUS**

Output

- VOUT+ → **5V RAIL (Power PCB)**
- VOUT- → **GND BUS**

Capacitors (right after converter output)

- **C5V_BULK**: 470μF–1000μF electrolytic across **5V RAIL ↔ GND BUS**

5V outputs (x3)

- **5V_OUT1+** → **Main PCB 5V**
- **5V_OUT2+** → **IR PCB 5V**
- **5V_OUT3+** → **Line PCB 5V**
- **5V_OUT4+** → **Ultrasonic PCB 5V**
- **5V_OUT5+** → **Open MV H7 R2 camera 5V**
- All their grounds → **GND BUS** (or provide 5V_OUTx- pins tied to GND BUS)

2) Step-Down Converter (3.3V)

Input

- VIN+ → **SW_12V+**
- VIN- → **GND BUS**

Output

- VOUT+ → **3.3V RAIL (Power PCB)**
- VOUT- → **GND BUS**

Capacitors (right after converter output)

- **C3V3_BULK**: 470µF–1000µF electrolytic across **3.3V RAIL ↔ GND BUS**

3.3V outputs (x3)

- **3V3_OUT1+** → **Main PCB 3.3V**
- **3V3_OUT2+** → **IR PCB 3.3V**
- **3V3_OUT3+** → **Line PCB 3.3V**
- Grounds → **GND BUS**

3) Step-Up Converter (48V) (for solenoid supply)

Input

- VIN+ → **SW_12V+**
- VIN- → **GND BUS**

Output

- VOUT+ → **48V RAIL**
- VOUT- → **GND BUS** (*shared ground unless you are intentionally isolating—most builds keep common ground*)

Capacitors (right after converter output)

- **C48V_BULK:** (typical) 100 μ F–470 μ F electrolytic rated $\geq 63V$ across **48V RAIL** $\leftrightarrow$ **GND BUS**

48V output (x1)

- **48V_OUT+** $\rightarrow$ **Solenoid Relay Module / Solenoid supply**
 - **48V_OUT-** $\rightarrow$ **GND BUS**
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12V output section

12V Out x1 (to Main PCB)

- **12V_OUT+** $\rightarrow$ **SW_12V+**
- **12V_OUT-** $\rightarrow$ **GND BUS**

Capacitor before/at 12V out (for motor noise)

- **C12V_BULK:** 1000 μ F–2200 μ F electrolytic ($\geq 25V$) across **SW_12V+** $\leftrightarrow$ **GND BUS**, placed near the 12V output connector
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Battery Capacity Indicator (after switch, before converters)

(So it turns off with the robot and reads the switched battery voltage)

- Indicator **+ / V** $\rightarrow$ **SW_12V+**
 - Indicator **GND** $\rightarrow$ **GND BUS**
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